

```
import numpy as np
```

```
Seasons = ["2010", "2011", "2012", "2013", "2014", "2015", "2016", "2017", "2018", "2019"]
```

```
Sdict = {"2010":0, "2011":1, "2012":2, "2013":3, "2014":4, "2015":5, "2016":6, "2017":7, "2018":8, "2019":9}
```

```
Sachin_Salary = [15946875,17718750,19490625,21262500,23034375,24806250,25244493,27849149,30453805,23500000]
Rahul_Salary = [12000000,12744189,13488377,14232567,14976754,16324500,18038573,19752645,21466718,23180790]
Smith_Salary = [4621800,5828090,13041250,14410581,15779912,14500000,16022500,17545000,19067500,20644400]
Sami_Salary = [3713640,4694041,13041250,14410581,15779912,17149243,18518574,19450000,22407474,22458000]
Pollard_Salary = [4493160,4806720,6061274,13758000,15202590,16647180,18091770,19536360,20513178,21436271]
Morris_Salary = [3348000,4235220,12455000,14410581,15779912,14500000,16022500,17545000,19067500,20644400]
Samson_Salary = [3144240,3380160,3615960,4574189,13520500,14940153,16359805,17779458,18668431,20068563]
Dhoni_Salary = [0,0,4171200,4484040,4796880,6053663,15506632,16669630,17832627,18995624]
Kohli_Salary = [0,0,0,4822800,5184480,5546160,6993708,16402500,17632688,18862875]
Sky_Salary = [3031920,3841443,13041250,14410581,15779912,14200000,15691000,17182000,18673000,15000000]
#Matrix
```

```
Salary= np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Pollard_Salary, Morris_Salary, Samson_Salary, Dhoni_S
Salary
```

```
array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
        25244493, 27849149, 30453805, 23500000],
       [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
        18038573, 19752645, 21466718, 23180790],
       [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
        16022500, 17545000, 19067500, 20644400],
       [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
        18518574, 19450000, 22407474, 22458000],
       [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
        18091770, 19536360, 20513178, 21436271],
       [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
        16022500, 17545000, 19067500, 20644400],
       [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
        16359805, 17779458, 18668431, 20068563],
       [    0,    0,  4171200,  4484040,  4796880,  6053663,
        15506632, 16669630, 17832627, 18995624],
       [    0,    0,    0,  4822800,  5184480,  5546160,
        6993708, 16402500, 17632688, 18862875],
       [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
        15691000, 17182000, 18673000, 15000000]])
```

Start coding or generate with AI.

```
-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_5177/1275587642.py in <cell line: 0>()
----> 1 salary

NameError: name 'salary' is not defined
```

Next steps: [Explain error](#)

```
Sachin_G = [80,77,82,82,73,82,58,78,6,35]
Rahul_G = [82,57,82,79,76,72,60,72,79,80]
Smith_G = [79,78,75,81,76,79,62,76,77,69]
Sami_G = [80,65,77,66,69,77,55,67,77,40]
Pollard_G = [82,82,82,79,82,78,54,76,71,41]
Morris_G = [70,69,67,77,70,77,57,74,79,44]
Samson_G = [78,64,80,78,45,80,60,70,62,82]
Dhoni_G = [35,35,80,74,82,78,66,81,81,27]
Kohli_G = [40,40,40,81,78,81,39,0,10,51]
Sky_G = [75,51,51,79,77,76,49,69,54,62]
```

```
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samson_G, Dhoni_G, Kohli_G, Sky_G])
```

Games

```
array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
       [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
```

```
[79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
[80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
[82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
[70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
[78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
[35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
[40, 40, 40, 81, 78, 81, 39, 0, 10, 51],
[75, 51, 51, 79, 77, 76, 49, 69, 54, 62]]])
```

```
Sachin_PTS = [2832,2430,2323,2201,1970,2078,1616,2133,83,782]
Rahul_PTS = [1653,1426,1779,1688,1619,1312,1129,1170,1245,1154]
Smith_PTS = [2478,2132,2250,2304,2258,2111,1683,2036,2089,1743]
Sami_PTS = [2122,1881,1978,1504,1943,1970,1245,1920,2112,966]
Pollard_PTS = [1292,1443,1695,1624,1503,1784,1113,1296,1297,646]
Morris_PTS = [1572,1561,1496,1746,1678,1438,1025,1232,1281,928]
Samson_PTS = [1258,1104,1684,1781,841,1268,1189,1186,1185,1564]
Dhoni_PTS = [903,903,1624,1871,2472,2161,1850,2280,2593,686]
Kohli_PTS = [597,597,597,1361,1619,2026,852,0,159,904]
Sky_PTS = [2040,1397,1254,2386,2045,1941,1082,1463,1028,1331]
#Matrix
```

```
np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morris_PTS, Samson_PTS, Dhoni_PTS, Kohli_PTS, Sky_PTS])
```

```
array([[2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133, 83, 782],
       [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154],
       [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743],
       [2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112, 966],
       [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297, 646],
       [1572, 1561, 1496, 1746, 1678, 1438, 1025, 1232, 1281, 928],
       [1258, 1104, 1684, 1781, 841, 1268, 1189, 1186, 1185, 1564],
       [ 903,  903, 1624, 1871, 2472, 2161, 1850, 2280, 2593, 686],
       [ 597,  597,  597, 1361, 1619, 2026, 852,  0, 159, 904],
       [2040, 1397, 1254, 2386, 2045, 1941, 1082, 1463, 1028, 1331]])
```

```
mydata=np.arange(0,20)
print(mydata)
```

```
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19]
```

```
np.reshape(mydata,(4,5))
```

```
array([[ 0,  1,  2,  3,  4],
       [ 5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14],
       [15, 16, 17, 18, 19]])
```

```
mydata
```

```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19])
```

```
MATR1=np.reshape(mydata,(5,4),order='c')
MATR1
```

```
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11],
       [12, 13, 14, 15],
       [16, 17, 18, 19]])
```

```
MATR1[4,3]
```

```
np.int64(19)
```

```
MATR1[3,3]
```

```
np.int64(15)
```

```
MATR1[-4,-3]
```

```
np.int64(5)
```

```
MATR2=np.reshape(mydata,(5,4),order='F')
MATR2
```

```
array([[ 0,  5, 10, 15],
       [ 1,  6, 11, 16],
       [ 2,  7, 12, 17],
       [ 3,  8, 13, 18],
       [ 4,  9, 14, 19]])
```

```
MATR1
```

```
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11],
       [12, 13, 14, 15],
       [16, 17, 18, 19]])
```

```
mydata
```

```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19])
```

```
MATR2[0:2]
```

```
array([[ 0,  5, 10, 15],
       [ 1,  6, 11, 16]])
```

```
MATR3=np.reshape(mydata,(5,4),order='A')
MATR3
```

```
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11],
       [12, 13, 14, 15],
       [16, 17, 18, 19]])
```

```
a1=['welcome','to','datascience']
a2=['required','hard','work']
a3=[1,2,3]
```

```
[a1,a2,a3]
```

```
[['welcome', 'to', 'datascience'], ['required', 'hard', 'work'], [1, 2, 3]]
```

```
np.array([a1,a2,a3])
```

```
array([[ 'welcome', 'to', 'datascience'],
       [ 'required', 'hard', 'work'],
       [ '1', '2', '3']], dtype='<U21')
```

```
dict1={'key':'val1','key2':'val2','key3':'val3'}
```

```
dict1['key2']
```

```
'val2'
```

```
dict2 = {'bang':2,'hyd':'we are hear', 'pune':True}
```

```
dict3 = {'Germany':'I have been here', 'France':2, 'Spain': True}
```

```
dict3['Germany']
```

```
'I have been here'
```

```
Pdict['Sachin']
```

```
-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_5177/4021091776.py in <cell line: 0>()
----> 1 Pdict['Sachin']

NameError: name 'Pdict' is not defined
```

Next steps: [Explain error](#)

```
Pdict = {"Sachin":0,"Rahul":1,"Smith":2,"Sami":3,"Pollard":4,"Morris":5,"Samson":6,"Dhoni":7,"Kohli":8,"Sky":9}
```

```
Players = ["Sachin","Rahul","Smith","Sami","Pollard","Morris","Samson","Dhoni","Kohli","Sky"]
```

```
Pdict['Sachin']
```

```
0
```

```
Games[Pdict['Rahul']]
```

```
array([82, 57, 82, 79, 76, 72, 60, 72, 79, 80])
```

```
Pdict['Sky']
```

```
9
```

```
Sdict['2019']
```

```
9
```

```
Salary[Pdict['Sky']][Sdict['2019']]
```

```
np.int64(15000000)
```

```
Salary[Pdict['Sky']][Sdict['2019']]
```

```
np.int64(15000000)
```

```
Salary
```

```
array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
        25244493, 27849149, 30453805, 23500000],
       [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
        18038573, 19752645, 21466718, 23180790],
       [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
        16022500, 17545000, 19067500, 20644400],
       [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
        18518574, 19450000, 22407474, 22458000],
       [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
        18091770, 19536360, 20513178, 21436271],
       [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
        16022500, 17545000, 19067500, 20644400],
       [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
        16359805, 17779458, 18668431, 20068563],
       [      0,      0,  4171200,  4484040,  4796880,  6053663,
        15506632, 16669630, 17832627, 18995624],
       [      0,      0,      0,  4822800,  5184480,  5546160,
        6993708, 16402500, 17632688, 18862875],
       [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
        15691000, 17182000, 18673000, 15000000]])
```

```
Salary/Games
```

```
/tmp/ipykernel_5177/3709746658.py:1: RuntimeWarning: divide by zero encountered in divide
Salary/Games
array([[ 199335.9375,   230113.63636364,   237690.54878049,
        259298.7804878,   315539.38356164,   302515.24390244,
        435249.87931034,   357040.37179487,   5075634.16666667,
        671428.57142857],
       [ 146341.46341463,   223582.26315789,   164492.40243902,
        180159.07594937,   197062.55263158,   226729.16666667,
        300642.88333333,   274342.29166667,   271730.60759494,
```

```

289759.875 ],
[ 58503.79746835, 74719.1025641, 173883.33333333,
177908.40740741, 207630.42105263, 183544.30379747,
258427.41935484, 230855.26315789, 247629.87012987,
299194.20289855],
[ 46420.5, 72216.01538462, 169366.88311688,
218342.13636364, 228694.37681159, 222717.44155844,
336701.34545455, 290298.50746269, 291006.15584416,
561450. ],
[ 54794.63414634, 58618.53658537, 73917.97560976,
174151.89873418, 185397.43902439, 213425.38461538,
335032.77777778, 257057.36842105, 288918. ],
522835.87804878],
[ 47828.57142857, 61380. , 185895.52238806,
187150.4025974, 225427.31428571, 188311.68831169,
281096.49122807, 237094.59459459, 241360.75949367,
469190.90909091],
[ 40310.76923077, 52815. , 45199.5 ,
58643.44871795, 300455.55555556, 186751.9125 ,
272663.41666667, 253992.25714286, 301103.72580645,
244738.57317073],
[ 0. , 0. , 52140. ,
60595.13513514, 58498.53658537, 77611.06410256,
234948.96969697, 205797.90123457, 220155.88888889,
703541.62962963],
[ 0. , 0. , 0. ,
59540.74074074, 66467.69230769, 68471.11111111,
179325.84615385, inf, 1763268.8 ,
369860.29411765],
[ 40425.6 , 75322.41176471, 255710.78431373,
182412.41772152, 204933.92207792, 186842.10526316,
320224.48979592, 249014.49275362, 345796.2962963 ,
241935.48387097]]))

```

```
np.round(Salary/Games)
```

```
/tmp/ipykernel_5177/2909567671.py:1: RuntimeWarning: divide by zero encountered in divide
```

```

np.round(Salary/Games)
array([[ 199336.,  230114.,  237691.,  259299.,  315539.,  302515.,
   435250.,  357040.,  5075634.,  671429.],
 [ 146341.,  223582.,  164492.,  180159.,  197063.,  226729.,
   300643.,  274342.,  271731.,  289760.],
 [ 58504.,  74719.,  173883.,  177908.,  207630.,  183544.,
   258427.,  230855.,  247630.,  299194.],
 [ 46420.,  72216.,  169367.,  218342.,  228694.,  222717.,
   336701.,  290299.,  291006.,  561450.],
 [ 54795.,  58619.,  73918.,  174152.,  185397.,  213425.,
   335033.,  257057.,  288918.,  522836.],
 [ 47829.,  61380.,  185896.,  187150.,  225427.,  188312.,
   281096.,  237095.,  241361.,  469191.],
 [ 40311.,  52815.,  45200.,  58643.,  300456.,  186752.,
   272663.,  253992.,  301104.,  244739.],
 [ 0., 0., 52140., 60595., 58499., 77611.,
   234949., 205798., 220156., 703542.],
 [ 0., 0., 0., 59541., 66468., 68471.,
   179326., inf, 1763269., 369860.],
 [ 40426., 75322., 255711., 182412., 204934., 186842.,
   320224., 249014., 345796., 241935.]])

```

```

import warnings
warnings.filterwarnings('ignore')

```

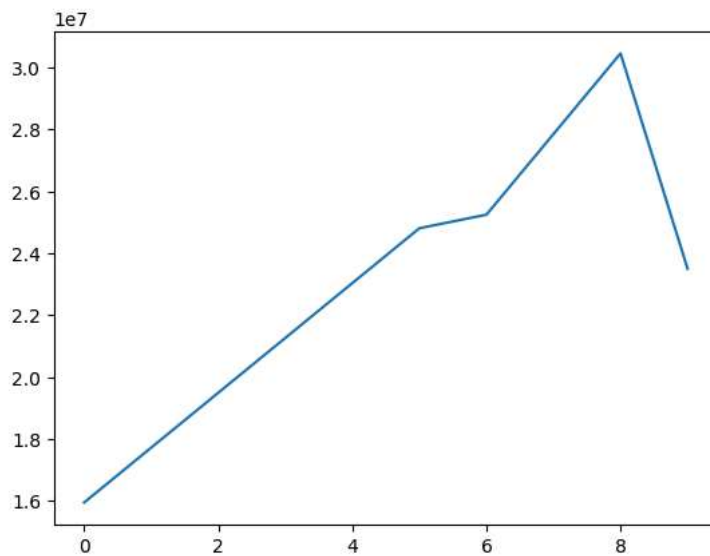
```

import numpy as np
import matplotlib.pyplot as plt

```

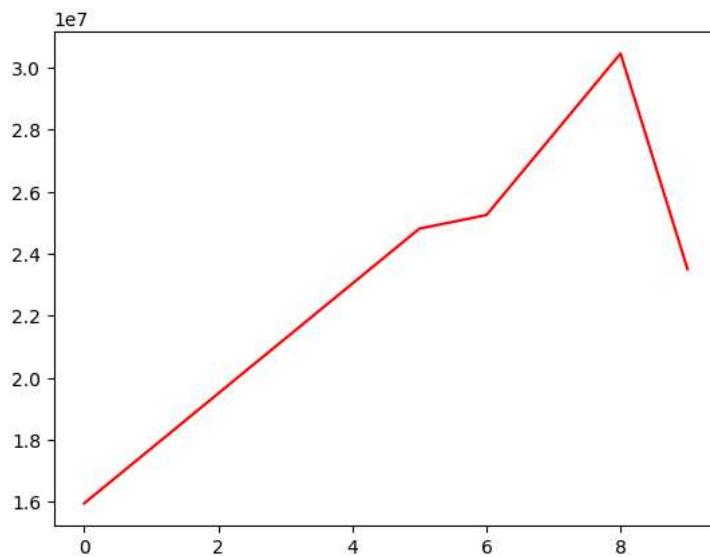
```
plt.plot(Salary[0])
```

```
[<matplotlib.lines.Line2D at 0x7b212e9e2030>]
```



```
plt.plot(Salary[0], c='red')
```

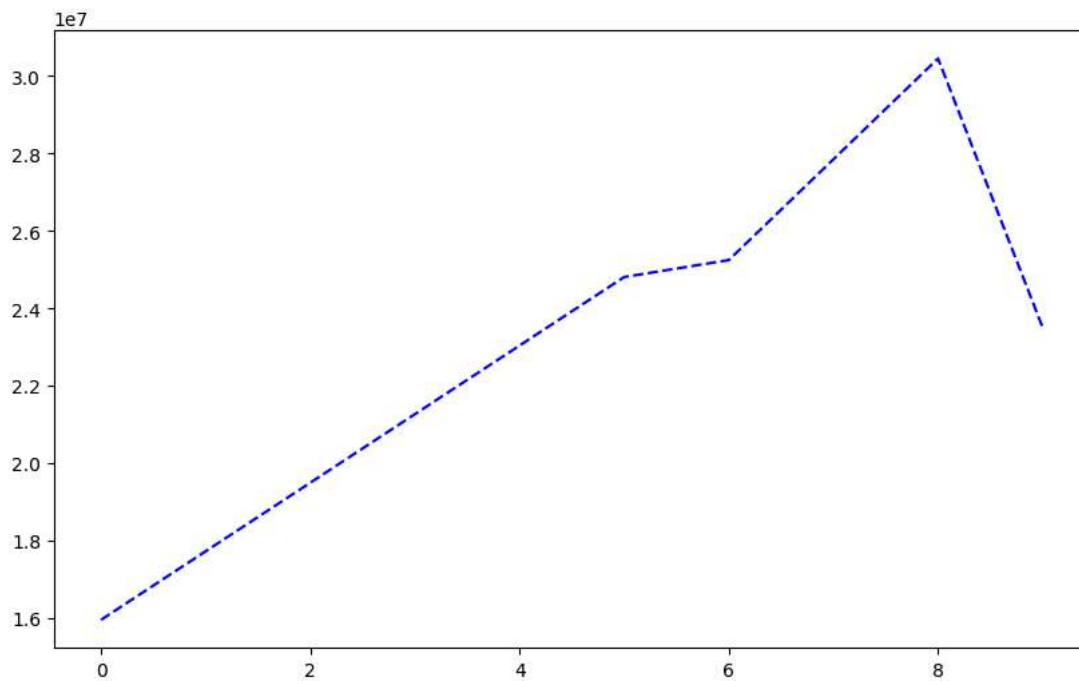
```
[<matplotlib.lines.Line2D at 0x7b21239719a0>]
```



```
%matplotlib inline  
plt.rcParams['figure.figsize']=10,6
```

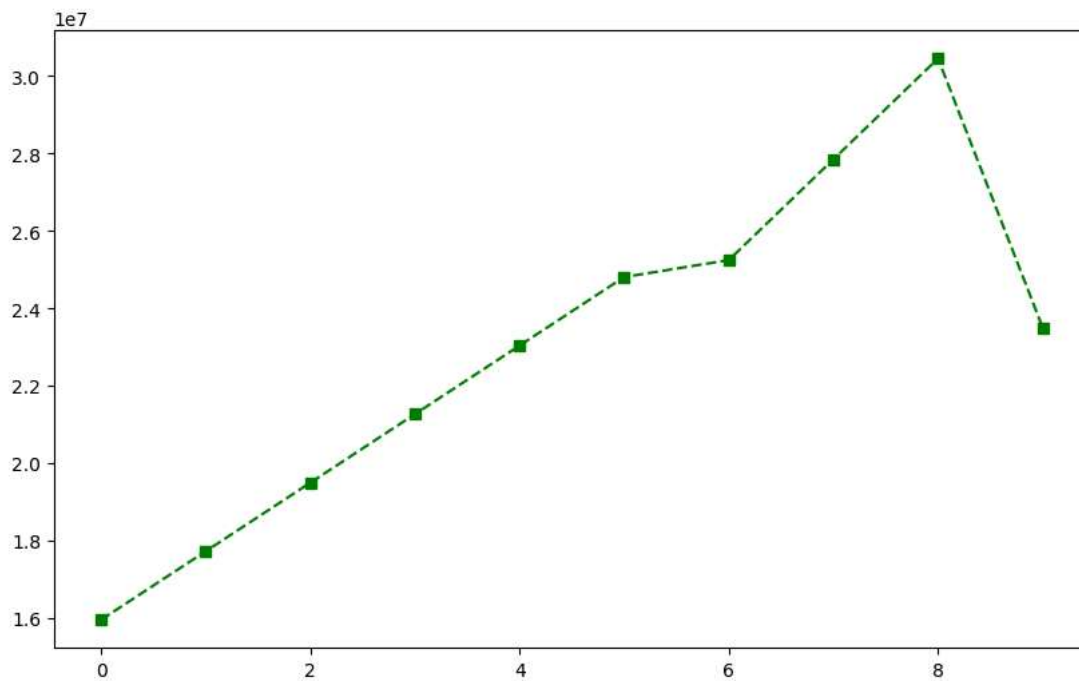
```
plt.plot(Salary[0],c='Blue',ls='dashed')
```

```
[<matplotlib.lines.Line2D at 0x7b2123ff92e0>]
```

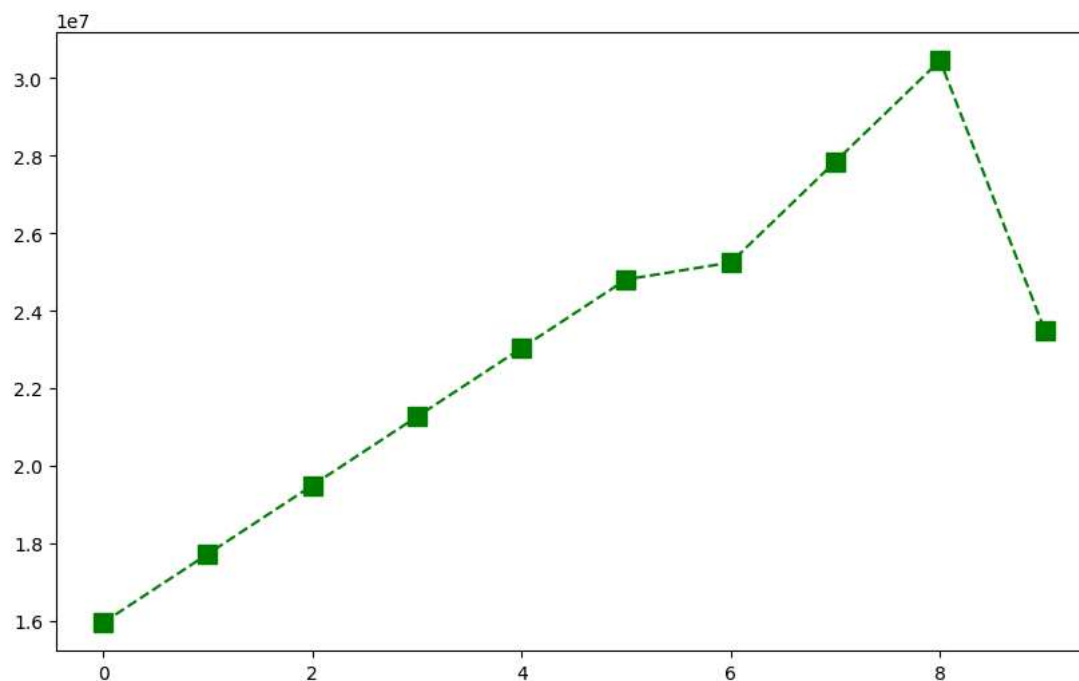


```
plt.plot(Salary[0], c='Green', ls = '--', marker = 's')
```

```
[<matplotlib.lines.Line2D at 0x7b2123f1b8c0>]
```



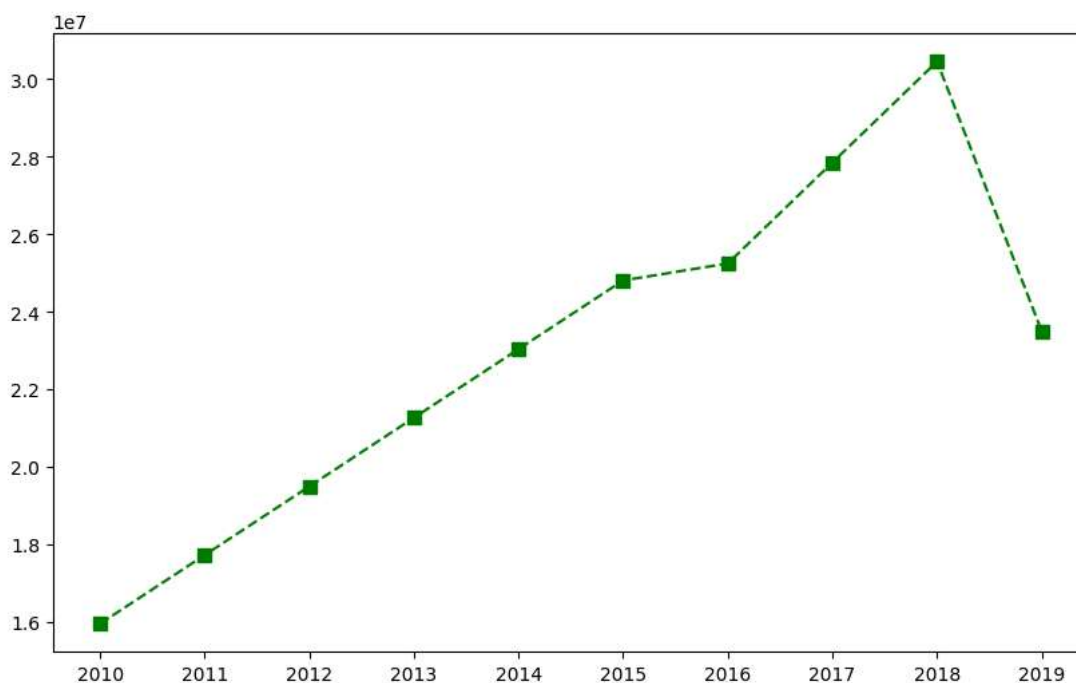
```
plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 10)  
plt.show()
```



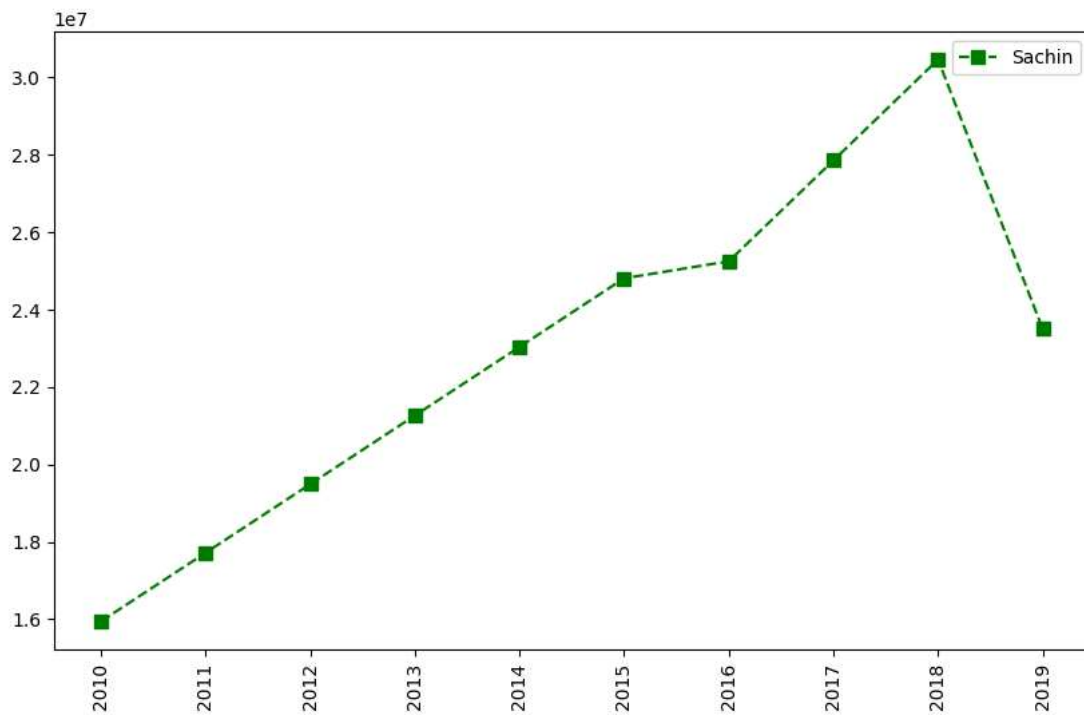
```
list(range(0,10))
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
plt.plot(Salary[0],c='Green',ls='--',marker='s',ms=7)
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



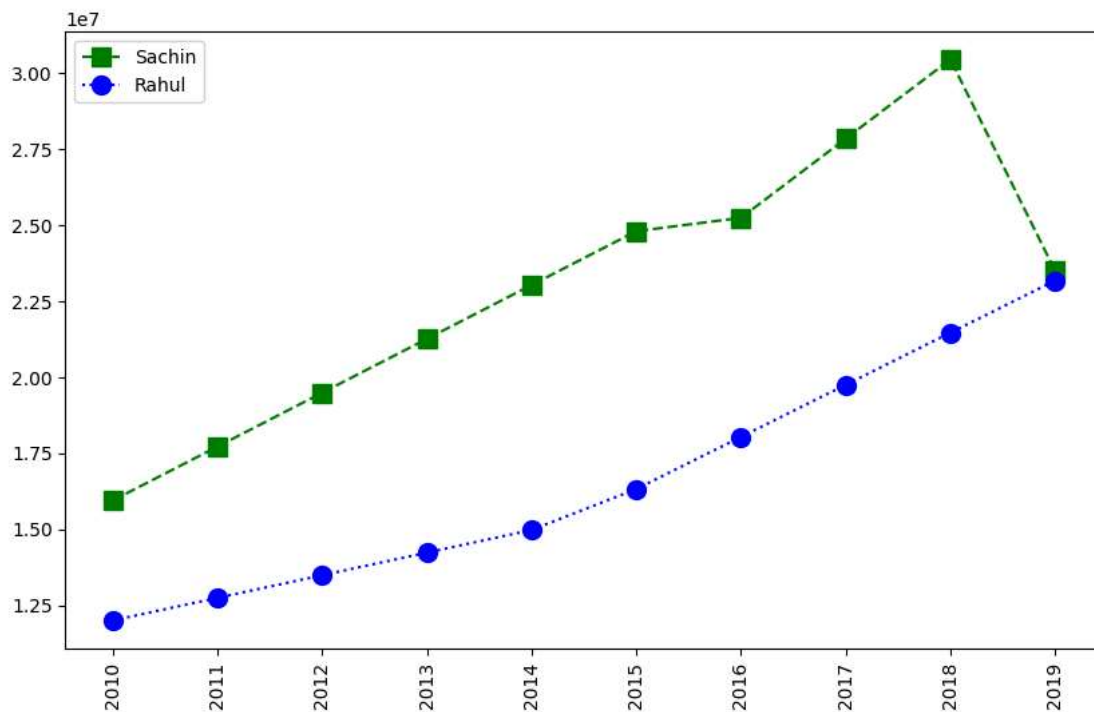
```
plt.plot(Salary[0],c='Green',ls='--',marker='s',ms=7,label=Players[0])
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')
plt.legend()
plt.show()
```

```
plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 10, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = ':', marker = 'o', ms = 10, label = Players[1])

plt.xticks(list(range(0,10)), Seasons,rotation='vertical')
plt.legend()

plt.show()
```



```
plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '--', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='purple', ls = '--', marker = '^', ms = 8, label = Players[2])

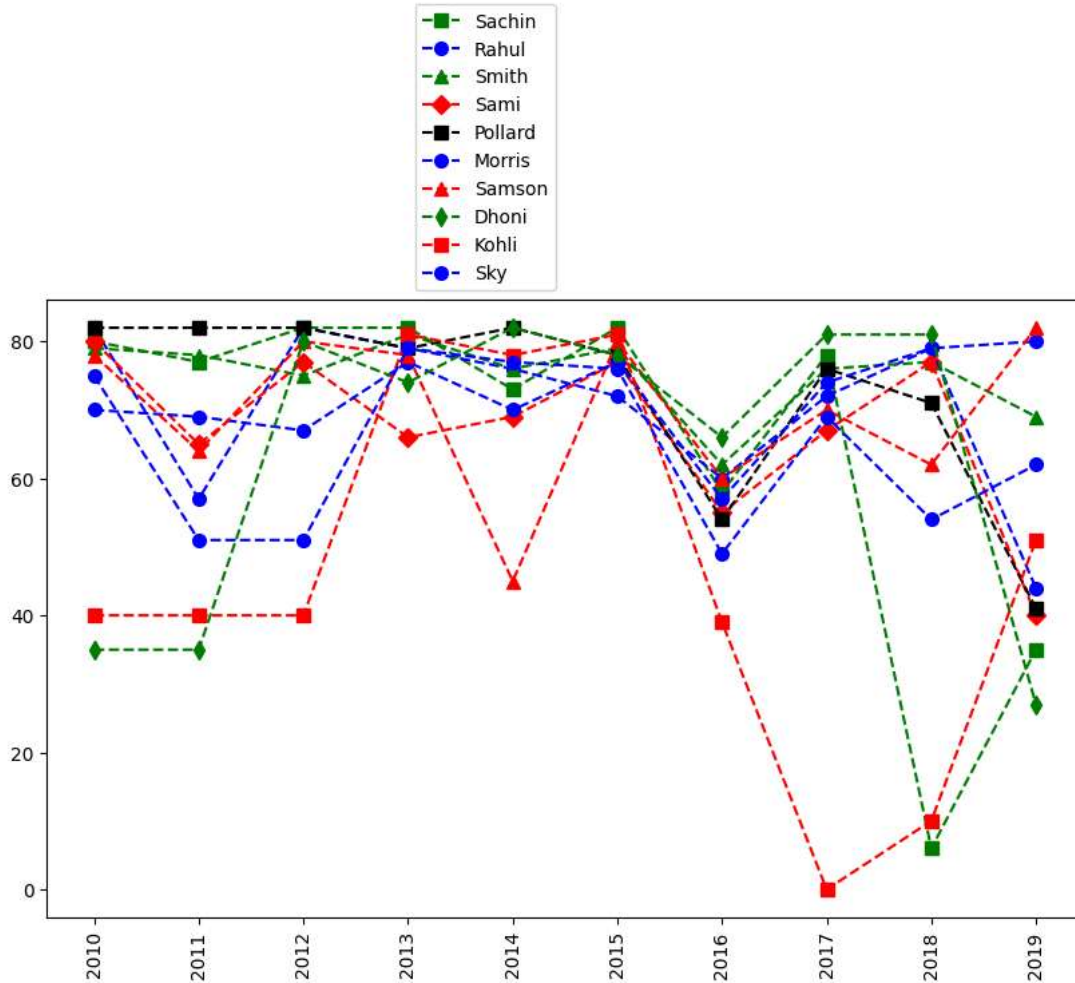
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()
```

```
plt.plot(Games[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Games[1], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[1])
plt.plot(Games[2], c='Green', ls = '--', marker = '^', ms = 7, label = Players[2])
plt.plot(Games[3], c='Red', ls = '--', marker = 'D', ms = 7, label = Players[3])
plt.plot(Games[4], c='Black', ls = '--', marker = 's', ms = 7, label = Players[4])
plt.plot(Games[5], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[5])
plt.plot(Games[6], c='red', ls = '--', marker = '^', ms = 7, label = Players[6])
plt.plot(Games[7], c='Green', ls = '--', marker = 'd', ms = 7, label = Players[7])
plt.plot(Games[8], c='Red', ls = '--', marker = 's', ms = 7, label = Players[8])
plt.plot(Games[9], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[9])

plt.legend(loc = 'lower right',bbox_to_anchor=(0.5,1) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()
```



```
plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[1])
plt.plot(Salary[2], c='Green', ls = '--', marker = '^', ms = 7, label = Players[2])
plt.plot(Salary[3], c='Purple', ls = '--', marker = 'D', ms = 7, label = Players[3])
plt.plot(Salary[4], c='Black', ls = '--', marker = 's', ms = 7, label = Players[4])
plt.plot(Salary[5], c='Red', ls = '--', marker = 'o', ms = 7, label = Players[5])
plt.plot(Salary[6], c='Red', ls = '--', marker = '^', ms = 7, label = Players[6])
plt.plot(Salary[7], c='Red', ls = '--', marker = 'd', ms = 7, label = Players[7])
plt.plot(Salary[8], c='Red', ls = '--', marker = 's', ms = 7, label = Players[8])
plt.plot(Salary[9], c='Red', ls = '--', marker = 'o', ms = 7, label = Players[9])

plt.legend(loc = 'lower right',bbox_to_anchor=(0.5,1) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()
```



